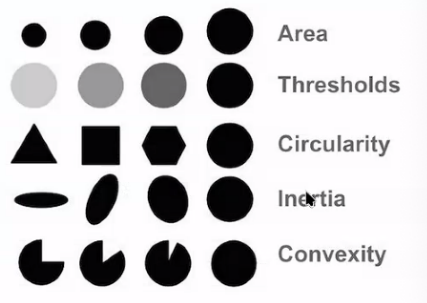


- OpenCV provides many filtering methods to isolate certain properties in an image
- Blurring and level curves can reduce noise
- `cv.findContours ()` can be used to filter based on advanced params

SimpleBlobDetector

- Makes a detector object when called

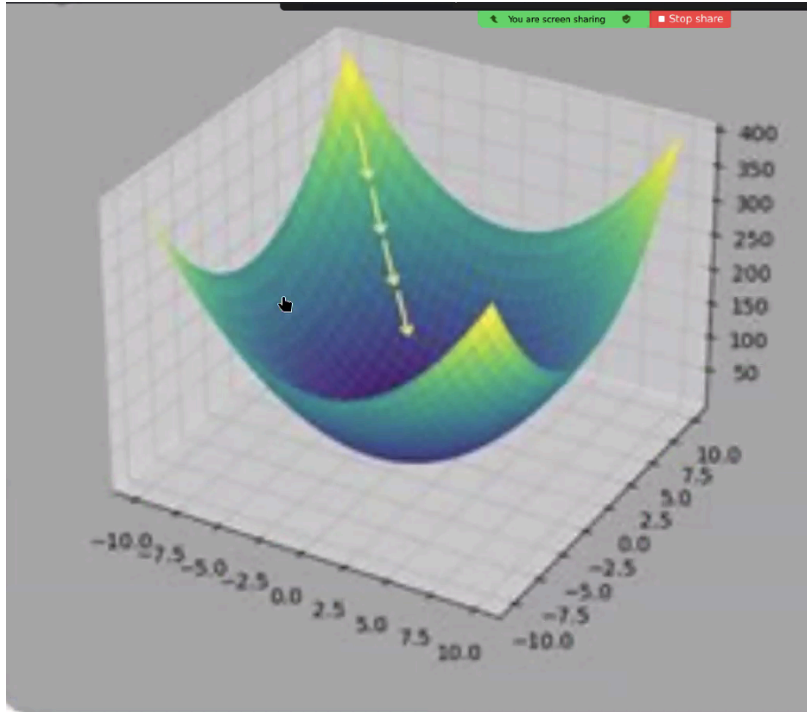


- Params in simpleBlobDetector
- A lot less versatile with non-bloby objects

Neural Network Basics

- Neural networks attempt to emulate thinking by making large graphs of simple functions
- Simple neural networks take in a set of inputs and have each node apply a weighted “activation function” to nodes in the previous layer
- Neural networks “learn” primarily by adjusting the weights of different nodes
- Start with a set of activation functions and weights for each node
- Loss function
- Keeping running a bunch of data through it with a bunch of parameters to decrease loss function
- Most data is reserved for training, the rest(smaller amount) is for testing
- Test loss > training loss because model is memorizing training data(overtraining)
 - Neural network becomes a storage of training data rather than predicting
- If we gave it similar data outside of the training dataset, it tests to see what the model predicts
- Overfitting happens if you train your model for too long
- reinforcement learning
 - Model learns on simulations
 - Infinite test data if you have robust simulations

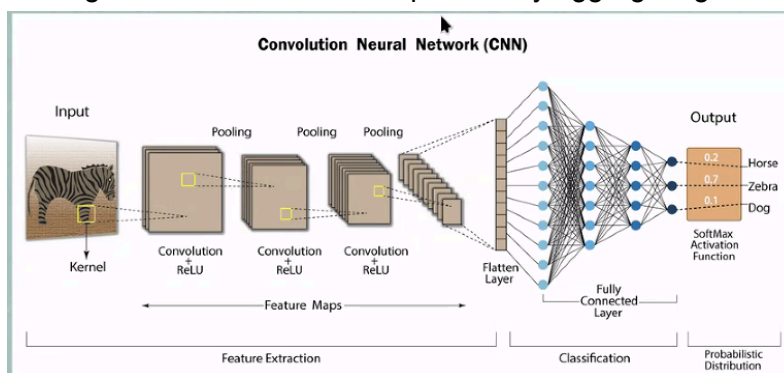
Gradient descent



- Graphing the loss of the model given a set of weights

Convolutional neural networks (CNNs)

- Convolution applies a kernel to an image to extract various features
- Pooling reduces the number of params by aggregating clusters of data into one value



How to train a model

- Collect data
 - Data needs to be robust and cover as many cases as possible.
- Annotate
 - More expensive algorithms like SAM can help with this
- Train
 - Different preprocessing and sampling methods may be optimal
 - Preprocessing data to make training easier for the model
 - Help the model separate features it needs

Skill Booster

1. OpenCV Blob Detection
2. YOLO